

ORF 524 – Final Exam

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Contents

1	Question 1: V-statistics and U-statistics (25 points)	1
2	Question 2: Uniform Inference at a Boundary (35 points)	3
3	Question 3: Large Sample Estimation and Inference (40 points)	6

Notes:

- Duration of examination: *4 hours*.
An additional 20 minutes are given to upload your solutions to Canvas.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Canvas.
- Only two pages of personal notes are allowed during the examination.
Basic auxiliary information about standard distributions is provided at the end of the exam.
This is otherwise a closed-book and individual examination.
Do not consult textbooks, online materials, or other sources.
Do not consult with classmates, other people or in online forums.

1 Question 1: V-statistics and U-statistics (25 points)

Let $(X_i : 1 \leq i \leq n)$ be an i.i.d. random sample with $\mathbb{E}[X_i] = \mu$ and $\mathbb{V}[X_i] = 1$. All limits are taken as $n \rightarrow \infty$, and $\rightsquigarrow$ denotes convergence in distribution.

A *V-statistic* is a generalization of the sample average to multiple sums over more than one observation at a time. For example, consider the following simple V-statistic:

$$\bar{V}_n = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n X_i X_j = \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2.$$

A *U-statistic* is a V-statistic which excludes the diagonal terms and is properly rescaled. In the above example, we have

$$\bar{U}_n = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n X_i X_j = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} X_i X_j, \quad \binom{n}{2} = \frac{n!}{2!(n-2)!}.$$

The *Hoeffding decomposition* gives a very useful representation for U-statistics (and hence also V-statistics). For the above example, it gives

$$\bar{U}_n = \mu^2 + \bar{L}_n + \bar{Q}_n,$$

where

$$\begin{aligned} \bar{L}_n &= \frac{1}{n} \sum_{i=1}^n L(X_i), & L(X_i) &= 2\mu(X_i - \mu), \\ \bar{Q}_n &= \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} Q(X_i, X_j), & Q(X_i, X_j) &= (X_i - \mu)(X_j - \mu). \end{aligned}$$

1. Show that $\mathbb{E}[\bar{L}_n] = 0$, $\mathbb{E}[\bar{Q}_n] = 0$ and $\mathbb{E}[\bar{L}_n \bar{Q}_n] = 0$.
Show that $\mathbb{V}[\bar{U}_n] = \mathbb{V}[\bar{L}_n] + \mathbb{V}[\bar{Q}_n]$.
2. Show that $\mathbb{V}[\bar{L}_n] = O(n^{-1})$ and $\mathbb{V}[\bar{Q}_n] = O(n^{-2})$.
Show that $\bar{L}_n = O_{\mathbb{P}}(n^{-1/2})$ and $\bar{Q}_n = O_{\mathbb{P}}(n^{-1})$.
3. Show that $\bar{V}_n = \bar{U}_n + O_{\mathbb{P}}(n^{-1})$.
4. Show that if $\mu \neq 0$ then $\sqrt{n}(\bar{U}_n - \mu^2) \rightsquigarrow \mathcal{N}(0, 4\mu^2)$.
5. Show that if $\mu = 0$ then $n\bar{U}_n \rightsquigarrow \chi_1^2 - 1$.

2 Question 2: Uniform Inference at a Boundary (35 points)

Let $(X_i : 1 \leq i \leq n)$ be an i.i.d. random sample from $X_i \sim \mathbb{P}_\theta \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$, where X_i is real-valued and $\Theta = [0, \infty)$. Suppose $\mathbb{E}_\theta[X_i] = \theta$ and $\mathbb{V}_\theta[X_i] = 1$ for all $\theta \in \Theta$. Consider the estimator:

$$\hat{\theta}_n = \max\{\bar{X}_n, 0\}.$$

1. Suppose $\mathbb{P}_\theta = \mathcal{N}(\theta, 1)$. Show that $\hat{\theta}_n$ is the maximum likelihood estimator for θ .
2. Show that $\hat{\theta}_n \rightarrow_{\mathbb{P}} \theta$ for all $\theta \in \Theta$.
3. Suppose $\theta > 0$. Show that $\sqrt{n}(\hat{\theta}_n - \theta) \rightsquigarrow \mathcal{N}(0, 1)$.
4. Suppose $\theta = 0$. Show that $\sqrt{n}(\hat{\theta}_n - \theta) \rightsquigarrow \max\{\mathcal{N}(0, 1), 0\} =: \mathcal{L}(0)$.
Give the exact form of the c.d.f. of the limiting law $\mathcal{L}(0)$.
5. Suppose $\mathbb{P}_\theta = \mathbb{P}_{\theta_n}$ with $\theta_n = \rho/\sqrt{n}$ for some $\rho > 0$.
Providing appropriate regularity conditions, show that $\sqrt{n}(\hat{\theta}_n - \theta_n) \rightsquigarrow \mathcal{L}(\rho)$.
Give the exact form of the c.d.f. of the limiting law $\mathcal{L}(\rho)$.
Hint: Recall the triangular array CLT.
6. Consider the following hypothesis testing problem for some $\theta_0 > 0$:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0.$$

The Wald statistic is $T_n(\theta_0) = \sqrt{n}(\hat{\theta}_n - \theta_0)$, which leads to the rejection rule (under H_0):

$$\text{Reject } H_0 \iff |T_n(\theta_0)| > \Phi^{-1}(1 - \alpha/2),$$

where $\Phi(x)$ denotes the c.d.f. of the standard Gaussian distribution, and $\alpha \in (0, 1)$.

- (a) Show that this test is a pointwise α -size test asymptotically. That is, show that

$$\sup_{\theta_0 > 0} \limsup_{n \rightarrow \infty} \mathbb{P}_{\theta_0} [|T_n(\theta_0)| > \Phi^{-1}(1 - \alpha/2)] = \alpha.$$

- (b) Show that this test is a uniform α -size test asymptotically. That is, show that

$$\limsup_{n \rightarrow \infty} \sup_{\theta_0 > 0} \mathbb{P}_{\theta_0} [|T_n(\theta_0)| > \Phi^{-1}(1 - \alpha/2)] = \alpha.$$

3 Question 3: Large Sample Estimation and Inference (40 points)

Let $(X_i : 1 \leq i \leq n)$ be i.i.d. with Lebesgue density $f(x - \theta)$ for some $\theta \in \Theta \subseteq \mathbb{R}$, where $f : \mathbb{R} \rightarrow \mathbb{R}$ is uniformly continuous and satisfies $f(x) = f(-x)$ and $f(0) > 0$. Suppose that n is odd and let $\hat{\theta}_{(1/2)}$ be the sample median. It was shown in one of the problem sets that

$$\sqrt{n}(\hat{\theta}_{(1/2)} - \theta) \rightsquigarrow N\left(0, \frac{1}{4f(0)^2}\right).$$

1. Show that the population median of X_i is unique and equal to θ .

Show that $\hat{\theta}_{(1/2)}$ is unique and consistent for θ .

Assuming that $\mathbb{E}[\hat{\theta}_{(1/2)}]$ exists, is $\hat{\theta}_{(1/2)}$ unbiased for θ ?

2. Assuming sufficient regularity conditions, find the log-likelihood, score and Hessian functions, denoted by $\ell_n(\theta)$, $\ell_{\theta,n}(\theta)$ and $\ell_{\theta\theta,n}(\theta)$, respectively.

Find the Cramér-Rao lower bound for unbiased estimators of θ , denoted by $\text{CRLB}(\theta)$.

Does $\text{CRLB}(\theta)$ depend on θ ?

3. Suppose that $\tilde{\theta}$ is a $\sqrt{n}$ -consistent estimator of θ . Consider the following one-step estimator:

$$\hat{\theta}_{\text{1step}} = \tilde{\theta} - \frac{\ell_{\theta,n}(\tilde{\theta})}{\ell_{\theta\theta,n}(\tilde{\theta})}$$

Providing appropriate regularity conditions, show that $\sqrt{n}(\hat{\theta}_{\text{1step}} - \theta)$ has an asymptotic normal distribution, and that it is asymptotically efficient.

Hint: Expand $\ell_{\theta,n}(\tilde{\theta})$ around $\ell_{\theta,n}(\theta)$.

4. Suppose that there exists a well-defined maximum likelihood estimator $\hat{\theta}_{\text{ML}}$ of θ . Providing appropriate regularity conditions, show that $\hat{\theta}_{\text{1step}}$ and $\hat{\theta}_{\text{ML}}$ are asymptotically equivalent, that is, $\sqrt{n}(\hat{\theta}_{\text{1step}} - \hat{\theta}_{\text{ML}}) = o_{\mathbb{P}}(1)$.

5. Consider now the special case $X_i \sim \text{Cauchy}(\theta, 1)$, which has Lebesgue density

$$f(x - \theta) = \frac{1}{\pi} \frac{1}{1 + (x - \theta)^2}.$$

- (a) Show that $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \sim \text{Cauchy}(\theta, 1)$.

Is $\bar{X}_n$ a consistent estimator of θ ?

- (b) Show that $\ell_{\theta,n}(\theta)$ has at least one root.

Show that if θ^* is a root of $\ell_{\theta,n}(\theta)$, then $\theta^* \in [\min_i X_i, \max_i X_i]$ almost surely.

- (c) Find the asymptotic efficiency of $\hat{\theta}_{\text{1step}}$ relative to $\hat{\theta}_{(1/2)}$.

Hint: for all $u > 0$,

$$J(u) := \int_{\mathbb{R}} \frac{1}{u + x^2} dx = \frac{\pi}{\sqrt{u}}.$$

(d) Let $\theta_0 > 0$ and consider the one-sided testing problem:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0.$$

Propose an asymptotic level $\alpha \in (0, 1)$ hypothesis test based on $\hat{\theta}_{1\text{step}}$.

Show that the power function of your test satisfies $\lim_{n \rightarrow \infty} \beta_n(\theta) = 1$ for all $\theta > \theta_0$.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$